

Explainer: Search

Duke OLLI Digital Explorers SIG

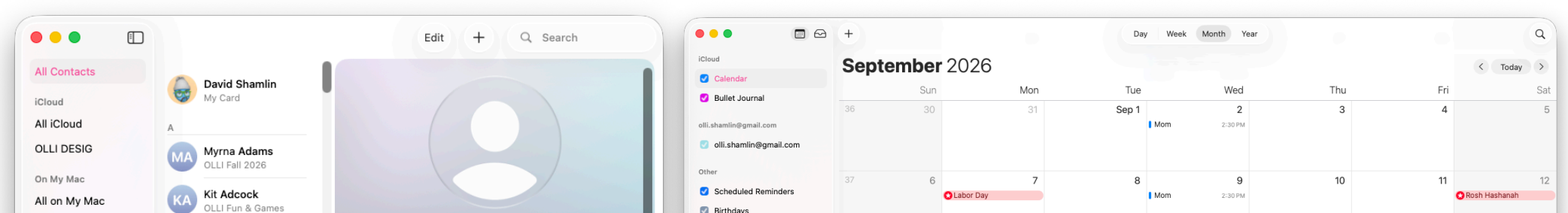
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General overview: These pages describe key concepts and terms related to the “search” and “find” controls/commands found in many app. In giving you more insight into how search technology works, my intent is to make it easier to use the various available search tools more effectively and navigate your way through the results of searches (which can sometimes be voluminous).	3-7
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Search: Overview

- The amount of data that can be stored on our devices' drives and the amount of information on the Internet are so large that it has become a near necessity for software vendors to include search features in their apps and services. E.g., we grumbled in some of previous sessions about struggling to find files we know exist on our drives or emails we previously read but didn't delete.
- Apps' search features (a.k.a. **search function**) are frequently represented in an app's UI with the magnifying glass icon. Sometimes this icon appears inside a textbox possibly along with the word "search". See the image below on the left; this is a screenshot of the Contacts app (on a Mac). In other cases, you click on the magnifying glass icon to cause a "search" textbox to appear. See the image below on the right; this is a screenshot of the Calendar app (on a Mac).



Search: Key Terms

- The piece of software that “drives” a **search function** is typically referred to as a **search engine**.
- Some app’s that have a **search function** have a **search engine** that is “built in” to the app. I.e., the app’s search function uses a search engine that is exclusively dedicated to that app.
- Other app’s **search functions** use a **search engine** that is a separate/independent piece of software. These “stand-alone” **search engines** can be used by multiple apps to implement their **search functions**.
- Some apps have a **find** feature/function (instead of or in addition to a **search function**). Apps that have a **find function** tend to be “document based” apps. I.e., apps that allow you to create/view/edit files (e.g., word-processing apps, spreadsheet apps, PDF viewers, etc). An example of an app that has a **find function** is the Pages app. To access the **find function** in the Pages app, you first have to have a document open in the Pages app; then...
 - On a Mac, click Edit > Find > Find... from the Menu Bar (or type CMD+F).

Note: if you have more than one Pages window open, the **find function** only searches the document in the active window.

 - On an iPad or iPhone, click the three dots in the toolbar, scroll down in the menu appears until you see the “Find” item and tap it.
- **Note:** The terms “search” and “find” tend to be used synonymously in UIs. There was a period of time in the digital technology’s past when “find” and “search” referred to fairly different capabilities; not so much these days. Apple’s Mail app is the only app I know of that has a **find function** and a **search function**; the Mail app’s **find function** searches for text in the currently selected email while the **search function** searches across all email messages in the selected/active mailbox.

Search: Key Terms

- The set of all possible things (i.e., pieces/body of data) that an app's search feature scans when performing a search is called the **search space**. For example, when you use the search feature in...
 - The Contacts app, the items you have created/stored (i.e., people) in your Contacts app is the search space.
 - The Calendar app, the events/appointments you have added in your calendar app is the search space.
 - The Preview app, the information in the document you are viewing is the search space.
- When a **search space** is massively large (e.g., all the files stored on your drives; the internet), **search engine's** tend to use an **index** to keep the amount of time it takes to complete an individual search as short as possible. These **indexes** (aka, **indices**) are analogous to the indexes found at the back/end of books and **search engines** use them in a similar way; they differ in that instead of associating page numbers with search terms, **search engine indexes** associate the search **terms** with the specific locations of possible terms. I.e., instead of searching through the data directly, the **search engine** consults the **index** for the **terms** of a **query** to determine which items in the **search space** match the **search query**.
- When you perform a search, the set of things the search feature uses is called the **scope** of the search. A search's scope can be the same as the **search space**. Some search features include the ability to reduce the number of things in the **search space** that are included in the **scope** of a search.
- The words or phrases you enter into a search textbox are referred to as search **terms**.
- A condition used to restrict the **scope** of a search to a subset of the **search space** is
 - E.g., search amazon for "men's shoes"; in left hand sidebar there are controls to limit the search to specific shoe sizes and widths.
- Conditions you impose to constrain the scope of a search are called **filters**.
 - The terms and filters used when searching are collectively referred to as the search **query**.
- The items returned by a search are referred to as **hits** or **matches**.
 - E.g., each of the URLs/pagesgoogle search...is called a **hit** or a **match**

Search engine strategies

Computer scientists have developed numerous search methodologies over the years. A given **search engine** will employ one or more of these methodologies when they perform a search. While there are many search methodologies, the ways search engines use them can be categorized into four *primary* strategies:

- **Keyword match**: This strategy involves scanning the **search space** for strings of characters that match a string of characters in a **query's terms**. E.g., given “cart” as the search **term**, a **search engine** that uses the keyword strategy will find two hits (both highlighted in red) in the following sentence: “I put a carton of eggs in the shopping cart.”
- **Fuzzy match**: This strategy provides tolerance for spelling variations and typos by identifying character strings that are approximately similar to the query **terms**. E.g., a fuzzy search for “librery” might treat “library” as a hit since the two words differ by only one character.

The two strategies above are typically employed when searching within document (or body of contiguous data) while the strategies below are typically employed when searching across multiple documents/files.

- **Best match** (or “rank by relevance”): This strategy is employed by **search engines** that cover massive **search spaces** (a prime example being Internet **search engines**) and that (typically) show you documents in the resulting list of **hits**. In such cases, the hits found by the **search engine** are shown in order of *estimated* relevance. I.e., the **search function** shows you the hits it decide have the highest degree of relevance first. **Search engines** that employ this strategy use various techniques to estimate a **hit's** degree of relevance.
- **Context-aware**: This strategy incorporates contextual factors into a **search engine's** behavior. Factors used can be previous **queries** you have performed (i.e., your search history), **hits** from previous searches you interacted with (i.e., clicked on), the type of device or app you use when performing a search (i.e., the same query executed on a laptop may produce a different set of **hits** than when it is executed on an iPhone; the same query executed with Safari may return different a different set of **hits** than when it is executed with Chrome.)
- **Hybrid**: Modern **search engines** rarely rely on a single strategy; instead they combine multiple strategies to leverage their respective strengths. Such **search engines** adapt the strategy they use to execute a particular **query** based on the structure of the **query** itself, the characteristics of items in the **search space**, and your behavior when you interact with the search engine and/or it's search results (i.e, **hits**).

Types of Search Spaces

Recall that **search functions** look for things within some **search space**. There are four kinds of general types of **search spaces**:

- A document

Document-based apps (i.e., apps that allow you to save/open files; e.g., word processing apps, spreadsheet apps, etc.) and apps that allow you to add/delete/edit items (e.g., Contacts, Calendar, Passwords) that have **find/search** functions fall into this category. When you perform a search/find in a document-based app, the document in the active app window is the **search space**. When you perform a search/find in apps like Contacts, Calendar, or password, the items you have created/saved in the app comprise the **search space**.

Find and search functions in document-based apps and apps that store items on your behalf (e.g., Contacts, Calendar, Passwords) have this type of general **search space**. **Search engines** with this kind of general **search space** tend to employ keyword match and/or fuzzy match strategies.

- A device

Most operating systems include a **search engine**. These kinds of **search engines** use the device's local drive and possibly attached cloud drives as their **search space**. Apple's OS **search engine** is called Spotlight. The Finder app on Macs and the Files app on iPhones & iPads have **search/find functions** that use Spotlight as well *to the best of my knowledge*. (Spotlight and Apple file manager's **search/find functions** are discussed in more detail later in this document.)

The **search functions** in the Finder and Files apps fall into this category. **Search engines** with this kind of general **search space** tend to employ keyword match, fuzzy match, and context-aware strategies and often in an adaptive hybrid way.

- A site

It has become common for sites on the internet to have a **search function**. Their **search space** is the content of the site itself. E.g., the **search space** of Wikipedia's **search function** is the set of pages on the site; the **search space** of Amazon's **search function** is the set of products Amazon sells. **Search engines** with this general type of **search space** tend to employ keyword match, fuzzy match, and context-aware strategies and often in an adaptive hybrid way.

- The Internet

- The **search space** of Internet **search engines** (like [Google](#), [DuckDuckGo](#), and [Microsoft Bing](#)) is the entirety of the surface web. (Recall our discussion of Internet layers during our Fall 2025 course; the image of the iceberg to the right visually represents these layers.)



Spotlight: Searching local & cloud drivers

- Spotlight is a search engine included with macOS, iOS, and iPadOS; you use it to search “system wide.” I.e., Spotlight’s search space includes all apps installed on your device, all files on your drives, the contents of *some* files stored on your drives, along with a number of other types of things (e.g., photos and text messages).
- Spotlight creates/maintains an index of all items and files on the system. It is designed to allow you to quickly locate a wide variety of items on your device including documents, picture, music, applications, and System Settings. In addition, specific words in documents and web pages in a web browser’s history or bookmarks can be searched. It also allows you to narrow down searches for files with filters using file creation dates, modification dates, sizes, file types, and other file attributes.
- Spotlight searches can find words inside files created by some apps.
- Spotlight searches can find words inside web pages in a browser’s history and bookmarks.
- Spotlight on iPhone and iPad includes fewer types of things in its search space than it does on Mac.
 - When Spotlight was first introduced on iPhones and iPads (four years after it was introduced on Mac), its **search space** was significantly smaller/less than that of Spotlight on Macs. since that time, Apple has expanded Spotlight’s **search space** on iPhone and iPad—although it appears to still be smaller than Spotlight’s **search space** on Mac. To the best of my current knowledge, Spotlight’s ability to find words/phrases inside documents on an iPad/iPhone is likely more limited than Spotlight’s ability to do that on a Mac.
- To invoke Spotlight...
 - On Mac: click the magnifying glass icon on the right hand side of the Menu Bar **or** use the `CMD+Space` keyboard shortcut.
 - On iPhone: tap the “search” textbox at the bottom of the Home Screen (on models with Face ID) or swipe down on the Home Screen or the Lock Screen.
 - On iPad: swipe down from the middle of the Home Screen or Lock Screen

Searching with the Finder and Files Apps

- The Finder and Files apps both have **search functions**.
 - With the Finder app, you can access the **search function** by (1) clicking on the magnifying glass icon/search text box found in the toolbar at the top of Finder windows, (2) clicking the “Find” item in the File menu, or (3) typing the keyboard shortcut CMD+F
 - With the Files app, look for the textbox with the label “Search” and/or the magnifying glass icon. (On my iPad and iPhone, they appear at the top of the screen.)
- Finder and Files apps’ search functions use Spotlight. I.e., these apps “pass” **queries** you enter in their search textboxes to Spotlight; Spotlight then executes the **query** and “passes” the **hits** back to Finder/Files; then Finder/Files show you the **hits** found by the search.
- Finder’s **search function** has UI controls for **filtering** search results that Files’ **search function** does not. Instead, Files’ search function *groups* search results by the type/kind of files/objects in search results.
- Help guide references
 - Finder: [Narrow your search results in Finder on Mac](#)
 - Files
 - On iPad: [Find and view files and folders in Files on iPad](#)
 - On iPhone: [Find and view files and folders in Files on iPhone](#)

Searching with the Mail App

- The Mail app on Mac has both a **search function** and a **find function**.
 - You access the search function by clicking on the magnifying glass icon or search textbox in the toolbar at the top of a Mail app window.
 - You access the find function by clicking `Edit > Find > Find...` from the Menu Bar or typing the shortcut `CMD+F`.
- The **search function** finds email messages that matches your search **query**. You can use it to search across all email accounts/mailboxes you have associated with the Mail app or filter on individual accounts/mailboxes.
- The **find function** scans the currently selected email message for the character string you enter into the “Find” textbox.
- The Mail apps **search function** has some **filtering** capabilities. How you access them varies between Mac, iPad, & iPhone.
 - On Mac and iPad, as you type in the “search” textbox, you will see filters that can be applied to your query appear in a popup menu beneath the “search” textbox.
 - On iPhone, you can apply filters to your query appear by tapping suggested filters found at the bottom of the query results. I.e., you must execute a query before the Mail app shows you a list of possible filters.
- Note: As I played/experimented with this **search function**
 - On iPhone, I experienced inconsistent behavior related with filters. I.e., sometimes it showed them to me; sometimes it didn't. I can't explain why. 😞
 - On iPad, when I selected a filter from the popup menu that appeared beneath the search box, I sometimes then saw controls appear in the sidebar (near the top; beneath the mailbox/account name). E.g., After I applied a date filter, buttons appear in the side bar that allow me to refine that filter to emails sent on that date, before that date, or after that date.

Searching the Internet

- Effectively using an internet search engine (like [google.com](https://www.google.com)) is arguably as much an art as it science. How you phrase your query can have a significant result on the results you get. The Google help page [at this link](#) further illuminates this point with a few examples.
- Google Search allows you to filter a search using what Google calls “operators.” E.g., you can restrict a Google search to only searching for pages the New York Times has published to their site by adding `site:nytimes.com` to your query. See [this Google help page](#) for more information.
- If/when you do internet searches to learn how to do something with your device or an app...
 - See if you can determine when a page was published or last updated; pages with “how to” info on digital technology sometimes do not age well. You will sometimes find this information on a page itself. Google’s search engine can provide some assistance as well; see [this help page](#) for more info.
 - A lot of people publish “how to” content on the internet.
 - Not all of those people know what they are talking about. When you find a page confusing/unhelpful, move on to another.
 - Some “how to” pages you find may have been published by software companies who are trying to market their product(s); these kinds of pages usually do provide some kind of “here’s how you can do it” information but then shift to promoting some software/app they claim will address your need even better. When I’m searching for “how to” answers and encounter pages like this, I consider them ads and move on to other hits in my search results.

References: Apple User Guides

- Mac
 - [Search for anything with Spotlight on Mac](#)
 - [Narrow your search results in Spotlight on Mac](#)
 - [Select result categories for Spotlight on Mac](#)
 - [Prevent Spotlight searches in specific folders or disks on Mac](#)
 - [Select result categories for Spotlight on Mac](#)
 - [If searching your Mac returns unexpected results](#)
 - [Spotlight settings on Mac](#)
 - [Take actions and shortcuts in Spotlight on Mac](#)
 - [Get calculations and conversions in Spotlight on Mac](#)
 - [Search your Clipboard history in Spotlight on Mac](#)
- iPhone: [Search with Spotlight on iPhone](#)
- iPad: [Search with Spotlight on iPad](#)
- Note: Somewhat interesting/curious to me that the iPhone User Guide has one page dedicated to Spotlight while the Mac User Guide has ten; why would Apple not include as much information on Spotlight in its iPhone User Guide as it does in its Mac User Guide? Regardless, here's a tip: there is a "search" textbox near the top of Apple User Guide pages; entering the word "search" into this textbox returns several pages of hits and in those hits I see links to pages that (based on the page titles) describe how to use the search functions in various Apple apps; consider reviewing/consulting these pages.

References: Apple Mail App Search & Find

- Mac: [Search for emails in Mail on Mac](#)
- iPhone: [Search for email in Mail on iPhone](#)
- iPad: [Search for email in Mail on iPad](#)
- Note: As I was crafting this page, I noticed that Apple publishes a help guide specifically for the Mail app on Mac; you can access that help guide via [this link](#). (If there are Mail app help guides for iPad and iPhone, I have not been able to find them.)

References: Google Search Help Guides

- [Google Search Help](#)

Note: The link above is to the “primary”/“main” Google Search “help center.” This help center site is organized differently than Apple’s user guides. Apple’s user guides are organized in a more “book-like” fashion (in that the pages do have an order to them and that order is shown in the help guides’ table of contents.) Google Search’s “help center” is a collection of pages with a very “loose”—if any—order; i.e., there is no clear/obvious “start here” page. This type of help site is sometimes referred to as a “knowledge base”; these types of help sites usually have their own search engine that tends to be helpful in discovering individual pages.

- [Learn search tips & how results relate to your search on Google](#)
- [Narrow your search results with filters](#)
- [Reliable results on Search](#)
- [Evaluate info you find with Google](#)
- [Learn more about a web page](#)
- [Learn more about an image](#)
- [Find videos in Search](#)